Request for Quotation (RFQ)

Compact SmartShoe PCB (v1.1)

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Confidential – For vendor quotation purposes only

1. Project Overview

We are seeking a professional partner to design and manufacture a compact, low-power electronic module for integration into smart footwear. The module will acquire IMU and barometric data from the GY-86 sensor board and stream processed outputs over BLE to a mobile application.

Key Objectives

• Compact, lightweight PCB with robust mounting for footwear integration.
• Reliable inertial and barometric sensing to enable cadence, stride length, and elevation estimation.
• BLE connectivity optimized for range and energy efficiency.
• Safe and simple magnetic charging solution with rugged pogo-pin interface.

2. Target Electrical Architecture

• **Power Module**: Adafruit PowerBoost 1000C (PID 2465) – integrated Li-ion charger with load sharing and 5 V boost output.
- Inputs: 3.7 V Li-ion battery (BAT) and external 5 V (USB-C or pogo) into USB pad.
- Output: regulated ~5.2 V from 5V pin to ESP32-S3 VBUS.
• **Microcontroller / Connectivity**: ESP32-S3 development board (5 V input, onboard 3.3 V regulator). Integrated BLE and Wi-Fi.
• **Sensors**: GY-86 module (MPU6050 + MS5611 + HMC5883L). Powered from the ESP32-S3 3.3 V rail.
• **Battery**: Single-cell Li-ion 3.7 V, 500–1000 mAh.
• **Protection**: Integrated pack protection or protected cell.

3. System Block Diagram

External 5 V (USB-C / Magnetic Pogo) ---> PowerBoost 1000C (USB input) Li-ion Cell 3.7 V ---------------------> PowerBoost 1000C (BAT input) | Integrated charger + boost v 5 V Output (~5.2 V) ---> ESP32-S3 Dev Board (VBUS) | Onboard 3.3 V Regulator (ESP32-S3) | 3.3 V Rail ---> GY-86 Sensor Board (I²C: SCL/SDA) Test Pads: 5 V, 3.3 V, VBAT, GND, SCL, SDA, TX, RX, RST/EN, BOOT, IMU INT Antenna Zone: ≥ 8 mm clearance, no copper/pour above or below antenna edge

4. Mechanical Requirements

• Board dimensions: ≤ 50 × 38 mm, FR-4 thickness 1.0–1.6 mm.
• Mounting: 4 × M2.5 holes at corners.
• Enclosure compatibility: Height ≤ 10 mm including tallest component.
• Environmental: Sweat and moisture resistant; conformal coating recommended.
• Antenna isolation: ≥ 8 mm clearance around ESP32 antenna.

5. Performance Targets

• IMU sampling: 200 Hz (minimum 100 Hz).
• Barometer sampling: 25–50 Hz.
• BLE throughput: ≥ 25 Hz streaming (orientation + step events).
• Battery life: ≥ 8 hours continuous operation with 800 mAh cell.
• Idle current: < 300 µA in deep sleep.

6. PCB & DFM Guidelines

• 4-layer stack-up recommended for RF.
• Controlled impedance (50 Ω for RF/USB).
• Continuous ground plane and stitching vias.
• TVS protection on connectors.

• ENIG finish; black soldermask.

7. Firmware Bring-Up

• Boot: USB enumeration and BLE advertising (`SmartShoe-XXXX`).
• BLE Services: custom GATT for orientation, cadence, steps, Δh.
• Sensor Fusion: complementary filter demo.
• Power Modes: deep sleep with wake-on-button.

8. Prototype Deliverables

• 5 assembled boards + 10 bare PCBs.
• Documentation: schematic, Gerbers, BOM, STEP, assembly drawings.
• Testing: power consumption, BLE range, functional jig.

9. Acceptance Criteria

• BLE range ≥ 8 m line-of-sight.
• Average current meets defined budget.
• Sensor performance within datasheet specifications.
• Safe magnetic charging at 5 V / 1 A.

10. Reference Components

• ESP32-S3-WROOM-1-N8R2
• GY-86 Sensor Module
• MAX17048 Fuel Gauge (optional)
• Adafruit PowerBoost 1000C (PID 2465)
• RGB 0603 LED

11. Compliance & Reliability

• Use RF-certified modules for FCC/CE compliance.
• Operating: 0–45 °C (charging), –10–50 °C (discharging).
• Humidity: ≤ 90% RH.
• RoHS compliant materials.

12. Commercial Terms

Please provide:
• NRE costs (design, layout, firmware, test jig).
• Per-unit pricing for 5, 25, and 100 units.
• Estimated lead times for design, fabrication, assembly, and delivery.

13. Vendor Clarifications Requested

1. Recommended ESP32-S3 antenna layout within ≤ 50 × 38 mm board.
2. Magnetic pogo-pin + magnet docking solution.
3. Power consumption estimates with chosen regulators.
4. Prototype build schedule and test methodology.

Requester: **Ratneswar Roychowdhury**

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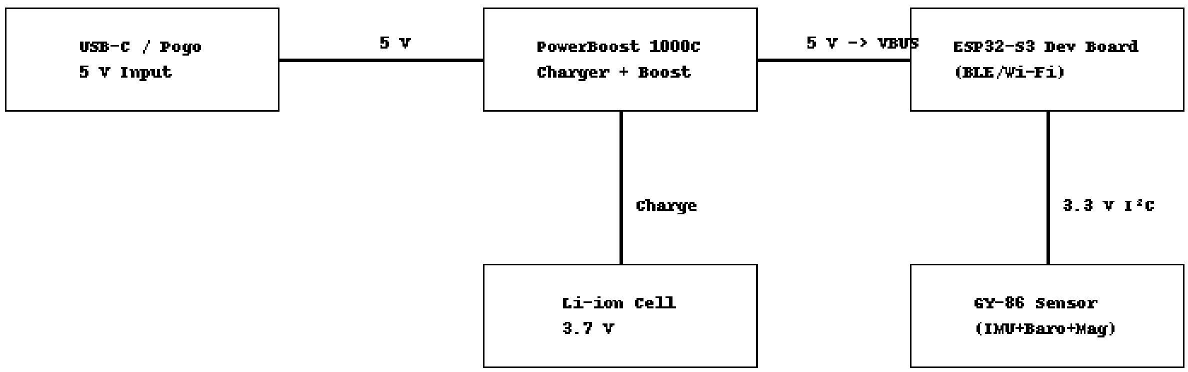

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